

Data Assisted Backscatter Communications using DECT-2020 NR+ as Ambient Signal

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Abstract—The Digital Enhanced Cordless Telecommunications (DECT-2020) New Radio (NR+), standardized by the European Telecommunications Standards Institute (ETSI), addresses International Telecommunication Union (ITU) IMT-2020 requirements for Internet of Things (IoT) and Industrial IoT. Although DECT-2020 NR+ offers robust connectivity, it does not support ultra-low power or batteryless operation. The 3rd Generation Partnership Project's (3GPP's) Ambient Internet of Things (AIoT) initiative targets ultra-low power IoT devices operating on harvested ambient energy, with backscatter communication as a promising technology. This paper integrates backscatter-based AIoT functionality into DECT networks, leveraging existing DECT transmissions and data-assisted channel estimation to recover backscattered information. An Ambient Backscatter Communication (AmBC) receiver is symbiotically embedded within the DECT receiver, using Received Signal Strength (RSS) of DECT data traffic. We open-sourced the Matlab code for the DECT receiver in a Software-Defined Radio (SDR) platform. Over-the-air experiments demonstrate feasibility, achieving Bit Error Rate (BER) performance as low as 0.02 under optimal conditions.

Index Terms—Ambient Backscatter Communications, Digital Enhanced Cordless Telecommunication, Received Signal Strength, Software-Defined Radio, Open-source Software

I. INTRODUCTION

The Digital Enhanced Cordless Telecommunications (DECT) standard was first established by the European Telecommunications Standards Institute (ETSI) in 1992 to provide short-range wireless audio communication. Since then, DECT has undergone several revisions aimed at achieving lower energy consumption and wider bandwidth. The latest specification DECT-2020 New Radio (NR+) [1] defines a new physical layer based on Cyclic Prefix Orthogonal Frequency-Division Multiplexing (CP-OFDM) modulation. This standard is specifically designed for the ultra-low energy Internet of Things (IoT) applications and massive Machine Type Communications (mMTC) in industrial automation [2].

Recently, the 3rd Generation Partnership Project (3GPP) initiated a study on the Ambient Internet of Things (AIoT), focusing on ultra-low power IoT devices that operate solely on energy harvested from ambient sources [3]. A promising technology for AIoT is backscatter communication, which

enables devices to transmit data by reflecting existing Radio Frequency (RF) signals without active transmission. The goal of 3GPP's AIoT work is to integrate these energy-harvesting devices into 6G mobile networks, facilitating sustainable and maintenance-free IoT deployments.

In this paper, we first propose integrating ultra-low power backscatter-based devices into DECT-2020 New Radio (NR) networks as a low-cost and energy-efficient alternative to the current 3GPP AIoT initiatives. While DECT-2020 NR+ offers robust connectivity suitable for many IoT applications, it may not inherently support the ultra-low power requirements essential for energy-autonomous AIoT devices [4]. By incorporating backscatter communication techniques, we aim to enable a new class of batteryless devices within the DECT-2020 NR+ framework, enhancing its applicability to energy-constrained AIoT scenarios.

Second, we adopt the Ambient Backscatter Communication (AmBC) approach for massive Machine Type Communication (mMTC) [5], wherein backscatter devices (BDs) utilize existing ambient RF signals, specifically DECT-2020 NR+ transmissions, as a means of communication. Unlike traditional Radio-Frequency Identification (RFID) systems that rely on a dedicated unmodulated carrier emitted by an RFID reader, AmBC devices modulate and reflect ambient signals already transmitted by other communication systems to convey information. While most AmBC implementations assume separate receivers for the ambient RF signal and the backscattered signal as proposed in the original work [6], we adopt the symbiotic radio paradigm [7] and propose the use of the existing DECT-2020 NR+ receiver for both mMTC and backscatter-based AIoT communications. The proposed system achieves high energy and spectral efficiency by leveraging the energy already emitted by the DECT-2020 NR+ system and reusing the same spectral resources for backscatter communication.

Finally, in the proposed system, the BD conveys information using Frequency Shift Keying (FSK). A modified DECT-2020 NR+ receiver employs data-assisted Received Signal Strength (RSS) to separate the BD modulated signal from natural multipath components in the Doppler-frequency domain. To validate the concept, we developed a proof-of-concept implementation and conducted over-the-air measurements. We open-sourced our Software-Defined Radio (SDR) Matlab DECT receiver on GitHub online [8]. The experimental results demonstrate the

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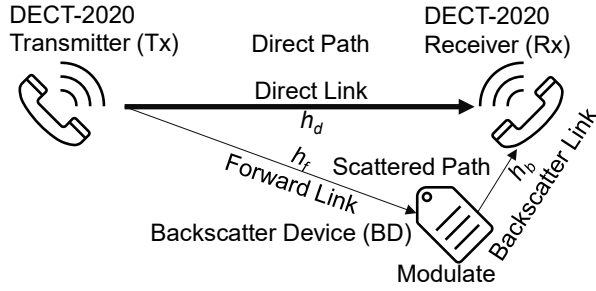


Fig. 1: An AmBC based on DECT-2020 NR+.

feasibility of the proposed approach, with the BD achieving a Bit Error Rate (BER) of 10^{-2} under the measured conditions.

The remainder of this paper is organized as follows. Section II presents the system model. In Section III, we implement the AmBC receiver based on DECT receiver. Section IV presents the measurement results. Finally, the last section concludes the paper.

II. SYSTEM MODEL

AmBC operates as a symbiotic communication system within the DECT infrastructure [7]. In this framework, DECT acts as the primary communication system, transmitting its own data while simultaneously providing the RF signals that illuminate the BD. The direct transmission between DECT nodes constitutes the direct path. Meanwhile, AmBC functions as the secondary communication system by modulating its own information onto the reflected DECT RF signals. This reflected transmission forms the scattered path. Notably, the DECT system is capable of receiving both its own signals and the scattered AmBC signals concurrently.

A. Channel Model

The channel model is depicted in Fig. 1. A transmitter (Tx) transmits a DECT signal s_t to a receiver (Rx) through two distinct paths: scattered path and direct path. The scattered path is composed by forward channel h_f and backscatter channel h_b . The direct path is denoted as h_d .

The overall system is implemented in an indoor scenario. The the large scale fading channels are expressed as

$$h_i = \frac{\lambda}{4\pi d_i} \exp\left(j \frac{2\pi d_i}{\lambda}\right), \quad i \in \{d, f, b\}, \quad (1)$$

where d_i , $i \in \{d, f, b\}$ refers to corresponding distances, i.e., distances of direct path d_d , forward path d_f , and backscatter path d_b . The direct and forward channels are either stationary or slow-moving, while the backscatter channel is stationary.

The overall Rx received signal is modeled as

$$s_r(t) = s_t(t) * h_d(\tau; t) + [(s_t(t) * h_f(\tau; t)) \cdot \rho(t)] * h_b(\tau; t) + w(t), \quad (2)$$

where $w(t) \sim \mathcal{CN}(0, \sigma_n^2)$ denotes Gaussian noise. The backscatter channel, $h_b(\tau; t)$, remains deterministic and flat-fading, effectively becoming a scalar multiplier $h_b(t)$. If BD is ideal, the BD either reflects ($\rho(t) = 1$) or absorbs ($\rho(t) = 0$)

all scattered signal power. Consequently, the signal model is simplified as

$$s_r(t) = s_t(t) * h(\tau; t) + w(t), \quad (3)$$

where overall channel $h(\tau; t)$ is either $h_{\text{off}}(\tau; t) = h_d(\tau; t)$ or $h_{\text{on}}(\tau; t) = h_d(\tau; t) + h_f(\tau; t)h_b(t)$. For further discussion convenience, (3) is normalized by σ_n . The channels are normalized with σ_n as well. Therefore, Signal-to-Noise Ratio (SNR) is defined as $\gamma = \mathbb{E}[\|h s_t\|^2]$.

The off state SNR γ_{off} is expressed as

$$\gamma_{\text{off}} = |h_r|^2 = \left(\frac{\lambda}{4\pi d_r \sigma_n}\right)^2, \quad (4)$$

and on state SNR γ_{on} is expressed as

$$\begin{aligned} \gamma_{\text{on}} &= |h_d + h_f h_b|^2 \\ &= \gamma_{\text{off}} \left[1 + \left(\frac{\lambda d_d}{4\pi d_f d_b}\right)^2 + \frac{\lambda d_d \cos[j2\pi(d_d - d_f - d_b)/\lambda]}{2\pi d_f d_b} \right]. \end{aligned} \quad (5)$$

The performance of AmBC is influenced by the geometric locations of TX, Rx and BD. This following inequality

$$\left(1 - \frac{\lambda d_d}{4\pi d_f d_b}\right)^2 \leq \frac{\gamma_{\text{on}}}{\gamma_{\text{off}}} \leq \left(1 + \frac{\lambda d_d}{4\pi d_f d_b}\right)^2 \quad (7)$$

indicates that the distance between BD and Rx—which can be more precisely defined as the wavelength number—impacts the backscatter SNR. In AmBC, it is desirable to maximize the difference between γ_{on} and γ_{off} . In the most severe situation, $\gamma_{\text{on}} = \gamma_{\text{off}}$, the backscatter states can not be disguised, rendering the AmBC system ineffective. According to (6), the BD is optimally positioned if $2(d_d - d_f - d_b)$ equals an integer multiple of the wavelength. Conversely, BD is at an ineffective location if $2(d_d - d_f - d_b)$ equals half of the wavelength.

B. DECT Frame Structure

The ETSI published the DECT-2020 NR+ specification physical layer in [1]. There are several configurations available for setting different bandwidths and slot durations. One frame, denoted as T_{frame} , consists of the $N_{\text{frame}}^{\text{frame}}$ OFDM symbols. Each OFDM symbol contains N_{sc} subcarriers. This physical resource is mapped to Resource Element (RE) $[k; l]$, where k denotes the subcarrier index and l denotes the position of OFDM symbol in the time domain. In this study, we focus on a Single-Input Single-Output (SISO) system. The received signal RE of (3) is rewritten as

$$S_r[k; l] = (H_d[k; l] + H_f[k; l]H_b[k; l])S_t[k; l] + W[k; l]. \quad (8)$$

The DECT transmission packet consists of Data Field (DF), Synchronization Training Field (STF) and Guard Interval (GI). The synchronization signal is consistently positioned at the beginning of each DECT packet on every fourth subcarrier, starting from the lowest occupied negative subcarrier and excluding the DC carrier, as illustrated in Fig. 2 red REs. Within a single frame, only N_{DF} OFDM symbols are allocated

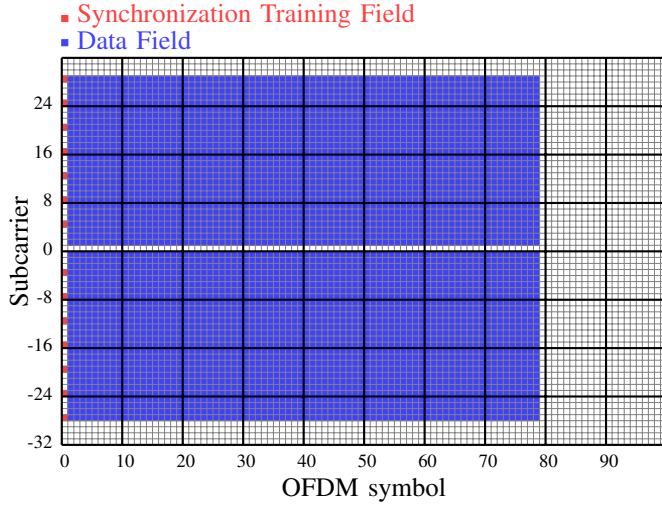


Fig. 2: Illustration of DECT-2020 resource mapping, STF and DF.

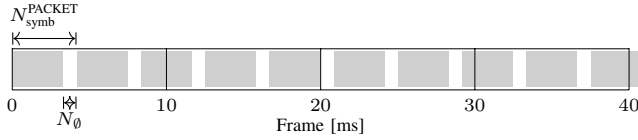


Fig. 3: Illustration of DECT frame structure and DF.

to the DF, which is beneficial for AmBC. The DF employs 4-QAM, providing an uniform symbol amplitude. The AmBC signal experiences periodic interrupts due to the DECT GI, as a period of $N_{\text{PACKET}}^{\text{symb}}$. As depicted in Fig. 3, these intervals last for N_0 OFDM symbol duration.

To address the missing samples, various interpolation algorithms can be employed. If the packet exhibits a unique pattern, as depicted in Fig. 3, the nonuniform sampling theorem [9, Ch. 10.2.2] can be utilized. Nevertheless, the DF may exhibit burstiness depending on the data traffic load. In such unpredictable scenarios, the Metropolis-Hastings algorithm [10] becomes an applicable alternative. In this paper, however, we opted for the most conventional method, zero interpolation, despite the fact that it introduces aliasing.

III. AMBC SIGNAL PROCESSING

In this section, we discuss the technique demodulating the AmBC signal from DECT signal power.

A. BD State Estimation

The DECT OFDM symbol duration is less than channel coherence time. Study in [11] shows that the BD signal predominantly influences the first channel tap $h(0; t)$. The BD signal of interest exhibits a flat spectrum [11]. For a specific l -th OFDM symbol, $H_r[l]$, $H_f[l]$ and $H_b[l]$ are scalars. The received signal in the “on” state is expressed as

$$S_r[k; l] = (H_r[l] + H_f[l]H_b[l])S_t[k; l] + W[k; l], \quad (9)$$

or in the “off” state is expressed as

$$S_r[k; l] = H_r[l]S_t[k; l] + W[k; l]. \quad (10)$$

The RSS of the l -th OFDM is denoted as

$$y[l] = \frac{1}{N_{\text{sc}}} \sum_{k=0}^{N_{\text{sc}}-1} |S_r[k; l]|^2. \quad (11)$$

According to study [12], the random variable y follows a *non-central chi-square distribution with $2N_{\text{sc}}$ degrees of freedom and noncentrality parameter $N_{\text{sc}}\gamma$* . This noncentral chi-square distributed random variable leads to a highly complex optimized receiver, as derived in [12, Eq. (62)]. Consequently, for simplification, y is approximated with a Gaussian distribution, which is expressed as

$$y \sim \mathcal{N} \left[1 + \gamma, \frac{1 + 2\gamma}{N_{\text{sc}}} \right]. \quad (12)$$

Mean and variance of y depend on backscatter state, respectively γ_{on} or γ_{off} . Consequently, the random variable y follows different Gaussian distribution based BD state: $y_{\text{on}} \sim \mathcal{N}(\mu_{\text{on}}, \sigma_{\text{on}}^2)$ or $y_{\text{off}} \sim \mathcal{N}(\mu_{\text{off}}, \sigma_{\text{off}}^2)$.

B. Square-wave FSK Symbol Detection

As previously discussed, the channel remains stable during one OFDM symbol. However, the channel experiences changes over longer time scales due to BD modulation. The RSS is estimated in each OFDM symbol. Thus the sampling rate is the OFDM symbol rate, i.e., $f_s = N_{\text{frame}}^{\text{frame}}/T_{\text{frame}}$. The BD only has two states on and off. The BD signal for FSK symbol i is expressed as

$$x_i[l] = \text{sgn}[\cos(2\pi l f_i / f_s)], i \in \{0, 1\}, \quad (13)$$

where $\text{sgn}(\cdot)$ is sign function and f_0, f_1 are FSK frequencies. BD is on when $x_i[l]$ is positive and is off when negative. The FSK waveforms are two $N \times 1$ column vectors, $\mathbf{x}_0$ and $\mathbf{x}_1$. In this paper, we apply the matched filter reported in [13]. Define the AmBC square-wave FSK receiver as

$$(\mathbf{x}_0 - \mathbf{x}_1)^H \mathbf{y} \underset{\mathcal{H}_0}{\overset{\mathcal{H}_1}{>}} 0, \quad (14)$$

where $\mathcal{H}_i, i \in \{0, 1\}$ refers to hypothesis that symbol i is transmitted by the BD.

C. Receiver Signal Processing

Fig. 4 illustrates the DECT system in the left side and the AmBC system in the right side as a symbiotic communication system. In this paradigm, DECT, the primary communication system, is demodulated first. Subsequently, AmBC, the secondary communication system, relies on the RSS results obtained from the primary DECT system.

After receiving the RF signal, the DECT receiver down-converts it to the baseband. The receiver, aware of the synchronization signal beforehand, continuously correlates it with the received signal. The sample at the correlation peak marks the beginning of the DECT packet. Following this, OFDM symbols are demodulated with cycle-prefix alignment. The subsequent steps for DECT involve channel estimation and equalization. At this stage, the AmBC receiver diverges from the DECT process. The AmBC receiver is primarily

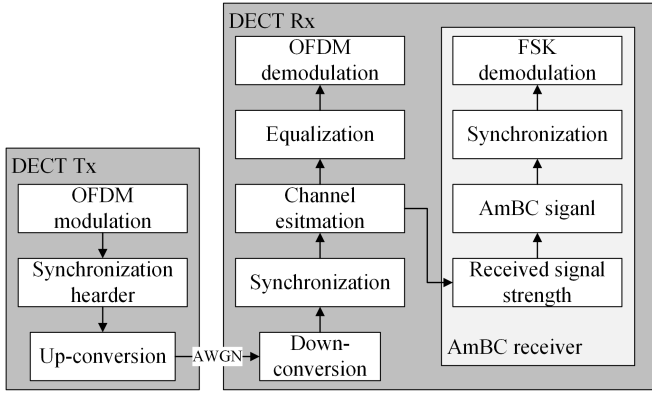


Fig. 4: Signal processing flow chart of DECT-2020 NR+ and AmBC.

concerned with the RSS of each OFDM symbol, which serves as “samples” for the AmBC signal.

The AmBC packet has its own synchronization header, consisting of three 7-bit Barker codes. The synchronization process for AmBC mirrors that of DECT. The start of sampling corresponds to the correlation peak with the known Barker code, thereby aligning the beginning of AmBC signal sampling. The AmBC square-wave FSK receiver then demodulates the AmBC information bits as shown in (14).

While the receivers for both AmBC and DECT operate in parallel, AmBC requires RSS data from the DECT receiver, forming a symbiotic relationship as depicted in Fig. 4. Without the DECT signal, AmBC cannot function independently, and therefore, the GI interrupts AmBC samples as well.

IV. MEASUREMENTS

The objective of this work is to demonstrate the feasibility of AmBC in the DECT system. For this purpose, an indoor over-the-air measurement campaign was conducted in this section.

A. Setup

To evaluate the feasibility of AmBC based on DECT, an over-the-air experiment was conducted in an indoor office environment, as depicted in Fig. 5. The Tx, Rx and BD were placed on a desk within a Line-of-Sight (LOS) propagation channel.

A Commercial Off-the-shelf (COTS) DECT platform, Nordic Semiconductor nRF91, transmitted DECT signal at a central frequency of 1888.704 MHz with a bandwidth of 1.728 MHz.

The BD comprised three parts, a specially designed RF front end, a Microcontroller Unit (MCU) and an antenna. The bespoke RF front end, which switches between two load impedances, has been previously studied in [14]. The antenna used an Anritsu Portable Antenna P/N 2000-1030, operational between 1.71 and 1.88 GHz. A Raspberry Pi nano RP2040 was embedded to provide clock signals and AmBC packet timing.

The AmBC packet consisted of two parts: a synchronization header and data. Symbols were modulated using square-wave FSK, with frequencies of 300 Hz and 625 Hz, and a symbol

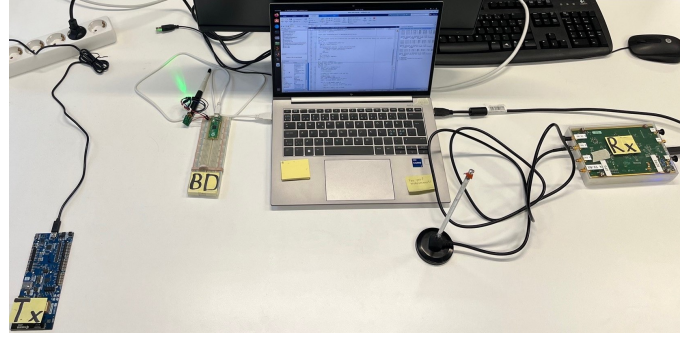


Fig. 5: Illustration of the measurement setup consisting of a commercial DECT-2020 Tx, a BD, and a SDR implementation of a DECT-2020 Rx.

duration of 20 ms. The total packet comprised 101 bits lasting 2.02 s. Between packets, a 0.1 ms idle interval was inserted to separate consecutive packets and enable header synchronization on receiver side. The BD operated independently without any knowledge of the DECT signal. Considering the clock drifting between Tx and Rx, the packet was deliberately designed to be short to prevent clock error accumulation and to ensure symbol demodulated within correct duration.

For DECT signal reception, we created a Matlab DECT toolbox, which we open-sourced on [8]. The toolbox implements DECT physical and MAC layers. The processing in Matlab is applied to I/Q samples acquired over the air by Universal Software Radio Peripheral (USRP) B210. The USRP has 12-bit Analog-to-Digital Converters (ADCs). The samples were collected by disabling Auto Gain Controller (AGC). The DECT receiver in Matlab synchronizes with DECT packets, decodes the data and re-encodes it for data assisted channel estimation. The channel estimates are fed to a separate AmBC receiver also implemented in Matlab.

B. Results

Fig. 6 illustrates the RSS variations over a longer period containing multiple frames. It exhibits slow changes due to channel slow fading as well as faster, discontinuous steps superposed on the slow fading, which are visible in Fig. 6. These steps are modulated by BD. The square-wave FSK was designed to operate at a frequency higher than natural Doppler frequency, allowing separation from slow changes in the channel attributed to natural Doppler. However, the BD state flipping frequency remains lower than the DECT OFDM symbol rate to avoid adversely affecting DECT performance.

Fig. 7 illustrates the measured BER of AmBC compared to the theoretical BER curve [15, Eq. (4.5-46)] expressed as

$$P_e = Q(\sqrt{\gamma_{\text{eff}}}), \quad (15)$$

where the x-axis is the normalized effective SNR of AmBC, denoted as γ_{eff} , and $Q(\cdot)$ is complement of the standard normal cumulative distribution function. It was observed that proposed AmBC achieved 0.02 BER. It was also found that the measured results align well with the theoretical curve. The short AmBC packet only contains 101 bits. Consequently, BER

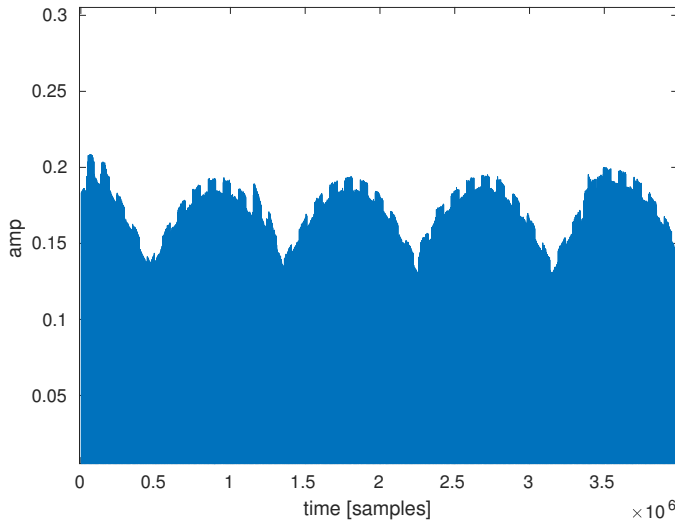


Fig. 6: Measured RSS for each DECT-2020 OFDM symbol. The square-wave pattern observed in the signal envelope results from the BD modulation.

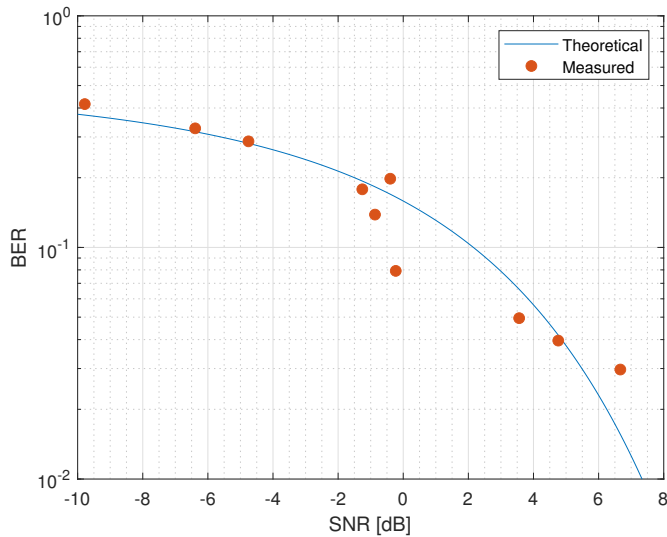


Fig. 7: BER performance of data-assisted DECT-2020-based AmBC receiver.

values lower than 10^{-2} were not included this measurement. Similarly, for packets without any error, BER is not applicable.

C. Discussion

According to the over-the-air measurement results in Fig. 7, some packets exhibited relatively poor performance with BERs exceeding 10^{-1} . To address such cases and improve overall system robustness, incorporating channel coding techniques—similar to those proposed in [16]—is a promising direction for future work.

V. CONCLUSIONS

In this paper, we explored the integration of ultra-low power backscatter-based devices into DECT-2020 NR+ networks in the AmBC framework. Additionally, we conducted over-the-air experiments to validate the feasibility of the proposed system. The measurement campaign was conducted in a real

indoor environment using a commercial DECT platform. Our measurement results found that a BER performance for the BD link as low as 0.02 was achieved under optimal conditions. The negligible power consumption of the RF diode switching in the BD, therefore AmBC aligns with the 6G vision of 3GPP in terms of ultra-low power communication. The acquired results demonstrate that AmBC effectively integrates with DECT, offering a supplementary communication method particularly suited for AIoT.

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